

Mahnoor Aetzaz

224-277-9299 | mahnooraetzaz@yahoo.com | [linkedin.com/in/mahnoor-aetzaz](https://www.linkedin.com/in/mahnoor-aetzaz) | github.com/mahnoor670 | mahnooraetzaz.vercel.app

EDUCATION

University of Illinois Urbana-Champaign

Bachelor's of Science, Computer Science & Education: Learning Sciences

GPA: 3.77 / 4.0

Champaign, IL

Expected May 2028

Relevant Coursework: Data Structures & Algorithms, Intro to Computer Systems, Discrete Structures, Linear Algebra, Statistics & Probability I, Database Systems, Applied Machine Learning

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, HTML/CSS, SQL, C, C++, C#

Web & Backend: React, Node.js, Flask, REST APIs, SQLite, SQLAlchemy

Dev Tools: Git, Docker, Linux, Visual Studio Code, Adobe Acrobat Pro

Data & ML: NumPy, Pandas, Matplotlib, SciPy, Jupyter, Google Colab, Gemini API

EXPERIENCE

Software Development Engineering Intern

Sept 2026 – Dec 2026

IDX Exchange

Remote

- Selected from a pool of **280+** applicants for a remote software engineering internship building production-style tools for real estate data platforms
- Develop full-stack features for a property search platform using React, Node/Express, and MySQL, working with real-world MLS datasets
- Design and implement RESTful APIs for property listing filtering, pagination, and search, applying Git workflows for collaborative development

Information Technology Computer Assistant (Hardware/Software)

Aug 2026 – Present

IT Partners, College of Education, University of Illinois Urbana-Champaign

Champaign, IL

- Provide first-line technical support for faculty, staff, and students, resolving **25-30+** hardware, software, and network tickets weekly through the department's ticketing system
- Set up, configure, and maintain institutional devices and systems, ensuring reliable day-to-day technology operations for the college

Linear Algebra with Computational Applications Course Assistant

Aug 2026 – Present

University of Illinois Urbana-Champaign

Champaign, IL

- Support instruction for Linear Algebra with Computational Applications (MATH 257), guiding sections of **100+** students through Jupyter-based Python labs during live Zoom and discussion sections
- Grade homework and host office hours for a course of **1,000+** students, providing individualized feedback, resolving code errors, and helping students connect linear algebra theory to Python.

Marketing Committee Co-Chair

Jan 2025 – Sep 2026

Reflections | Projections

Champaign, IL

- Led outreach strategy for the largest student-run tech conference in the Midwest, serving **1,000+** annual attendees
- Managed a committee across social media, graphics, and video content, delegating tasks and aligning campaigns to conference goals

Intro to Computer Science I Course Assistant Tutor

Jan 2025 – Aug 2025

University of Illinois Urbana-Champaign

Champaign, IL

- Tutored **1,000+** students in Java fundamentals via the course's online platform, covering problem-solving, debugging, and core programming concepts
- Conducted one-on-one code reviews for student Java applications, reinforcing object-oriented design and data structures

PROJECTS

Algorithm Storybook | *React, Flask, Python, REST API, SQLite, SQLAlchemy*

GitHub | 2026

- Built a full-stack web application with **6** interactive algorithm levels, architected a REST API in Flask with **12** endpoints connecting a React frontend to a Python backend with persistent data storage
- Engineered real-time game state management in React using hooks to track user inputs, eliminated search spaces, and compute optimal moves across binary search and sorting algorithm levels
- Designed a backend evaluation engine that validates user guesses against optimal algorithm logic and returns structured JSON feedback, enabling step-by-step computational reasoning for learners
- Designed a relational data model spanning **3** database tables connecting teachers, students, and scores, enabling teachers to create classes and track student progress in a live dashboard

Email Drafting Assistant | *Python, Flask, Gemini API, HTML/CSS, JavaScript*

GitHub | 2025

- Built a reusable template system with dynamic placeholder substitution across multiple tone-specific formatting styles and persistent draft storage using Flask and HTML/CSS
- Structured prompt context for the Gemini API by enforcing user-provided sender, recipient, and tone inputs, ensuring generated emails stayed grounded in user intent
- Implemented front-end tone selection features that passed structured user preferences to the LLM, controlling output style across formal, casual, and persuasive registers